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Predictive Analysis of Hospital Overall Ratings Using Healthcare Quality and Safety Indicators

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***Abstract*— Star ratings of hospitals are a composite summary of several dozens of measures of quality that stakeholders depend on, however, it is hard to understand how particular measures contribute to it. This paper explores to what extent seven national comparison groups of quality and safety indicators as well as four attributes of hospitals can predict star ratings of hospitals in the United States. After cleaning the publicly available Hospital Information dataset (4,812 records), 3,567 hospitals having a proper 1-5 star rating were kept. A composite Healthcare Quality-Safety Score was created from the indicator groups, and three class-balanced classifiers – multinomial Logistic Regression, Decision Tree and Random Forest were trained within a leakage-free preprocessing pipeline and tested on a stratified 20% hold-**
nd Random Forest showed
ted F1 of 0.678 and 0.673

for predicting the star rating [1]. Hu and Nerenz demonstrated that overall scores have very narrow distribution range and are not significantly related to specific disease-related quality measures in California and New York [2]. Locey et al. provided an open source tool for estimation of ratings with 99.6% accuracy six months prior to their public release using the CMS methodology and corresponding data [3]. The above-mentioned studies operate with precise numeric measures; we are going to examine the information capacity of coarse national-comparison categories.

Application of machine learning to healthcare. The reviews on artificial intelligence in medicine emphasize increase of non-image data usage and discuss issues such as lack of data and biases [4]. They also provide an overview of multimodal

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